# When Does an Additional View Become Harmful?

## Marginal Utility of Degraded Observations in Multi-View 3D Reconstruction

Robot Vision, Term Project Proposal
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**Research question.** Given a fixed set of valid views, at what degradation level does adding one geometrically valid observation become worse than excluding it?

### 1. Motivation

Recent feed-forward 3D reconstruction models can infer camera and geometry directly from multiple images, but real observations often violate clean-image assumptions. Motion blur, defocus, low light, occlusion, and other degradations can reduce pose and geometry accuracy. Prior work has already shown that blurred frames can hurt SfM/visual odometry, that low-quality frames can be filtered, and that outlier views can damage modern feed-forward reconstruction.

Therefore, simply showing that motion blur lowers reconstruction quality, or that blurred frames should sometimes be removed, would not be a sufficient contribution. We instead study a narrower counterfactual: for the same reconstruction context, is a degraded observation still better than having no observation at that viewpoint at all?

The key distinction is between absolute robustness and marginal utility. A model may perform worse as blur increases while the added view is still beneficial relative to dropping it. The scientifically interesting failure occurs when the additional view becomes actively harmful.

### 2. Measuring Marginal Utility

Let K_-i be a fixed context of observations that excludes target frame Fi, and let Fi(b) be the same target observation at the same nominal viewpoint with motion-blur severity b. For reconstruction error E, define

$$\boldsymbol{U_i(b\,|K_{-i}) = E(K_{-i}) \;-\; E(K_{-i} \cup \{F_i(b)\})}$$

| Utility | Interpretation |
|---|---|
| U > 0 | Adding the degraded view improves reconstruction |
| U ≈ 0 | The view adds negligible information |
| U < 0 | Reconstruction is better when the view is withheld |

The zero crossing b* where $U_i(b^* \mid K_{-i}) = 0$ is the primary transition of interest. It is not an intrinsic property of a frame: it depends on the scene, the other views in $K_{-i}$, the reconstruction backend, and the evaluation metric. We therefore repeat the same intervention across many target/context pairs and analyze the distribution of U and b*.

### 3. Controlled Experiment

1. Choose a clean scene, target observation $F_i$, and fixed context $K_{-i}$.
2. Run reconstruction on $K_{-i}$ to obtain the drop baseline $E(K_{-i})$.
3. Generate a blur sweep $F_i(b_0)$, $F_i(b_1)$, … while keeping scene and nominal viewpoint fixed.
4. Run reconstruction independently on $\boldsymbol{K_{-i} \cup \{F_i(b)\}}$ for each severity.
5. Compute $\boldsymbol{U_i(b\,|K_{-i})}$ for pose and geometry metrics.
6. Repeat across scenes, target views, contexts, and reconstruction backends.

Motion blur should be generated by exposure integration along a camera trajectory whenever possible. For pure rotation, the intermediate views can be related by homographies without depth. For translational motion, depth-aware rendering or warping is preferable because blur is spatially varying and depth dependent.

## 4. Reconstruction Backends and Data

The initial backends are VGGT-based and MASt3R-based reconstruction. They are treated as reconstruction systems, not as keyframe selectors to be audited. Both receive identical, explicitly specified image sets.

ETH3D can provide calibrated clean sequences and ground-truth pose/geometry for controlled synthetic interventions. RealX3D is an important complementary benchmark because it already evaluates modern reconstruction under physically realistic degradations; if practical, it can be used as external validation rather than as the source of the novelty claim.

## 5. Evaluation

We will keep several metrics rather than collapse reconstruction quality into one scalar. Pose metrics include ATE/RPE or pairwise translation and rotation error as appropriate to the backend. Geometry metrics include depth error, point-cloud precision/recall/F1, Chamfer distance, completeness, or point-to-surface error where ground truth permits.

Primary aggregate results will be (1) utility curves $U(b)$, (2) the fraction of trials with $U<0$ at each severity, (3) the distribution of zero-crossing severity $b^*$, and (4) how these quantities change with overlap, baseline, redundancy, texture, scene geometry, and backend.

## 6. Relation to Prior Work and Expected Contribution

RealX3D already benchmarks motion blur and other degradations in multi-view reconstruction. Recent work on VGGT shows emergent rejection of irrelevant/non-overlapping distractor views, while G-MASt3R-SfM explicitly prunes unreliable views. Classical SfM and visual-odometry work has also used blur-aware frame filtering. These papers establish that bad observations and blur can be harmful.

Our intended contribution is therefore more specific: to characterize the conditional marginal value of one geometrically valid observation as its degradation severity is increased continuously, using an explicit drop baseline. The target phenomenon is not merely performance degradation but a sign change from beneficial to harmful input.

**Main claim to test**

More visual information is not always better: under some contexts, a sufficiently degraded but geometrically valid observation may have negative marginal utility relative to withholding it.